PYTHON FOR PHYSICS / WEEK 1

Does a smaller time step improve your simulation?

25-35 minutes | NumPy + Matplotlib | No solution included

Model a capacitor discharging through a resistor. Test forward Euler accuracy, rather than simply producing a plausible curve.

PHYSICAL MODEL

R = 10,000 ohms C = 100 x 10^-6 F V(0) = 5.0 V tau = RC

dV/dt = -V/(RC) **V_exact(t) = V(0) exp(-t/tau)**

YOUR TASK

1. Write a function taking time step dt that returns arrays of times and simulated voltages from t = 0 through t = 5 tau. Use forward Euler:

   V[n+1] = V[n] + dt x (dV/dt evaluated at V[n])

2. Run it with dt = 0.20, 0.10, and 0.05 seconds.

3. Plot all three simulations and the exact solution on one labeled graph.

4. For each step size, print the maximum absolute error at its own times:

   E_max = max over n of |V[n] - V_exact(t[n])|

5. In one or two sentences, describe how halving dt changes the error.

PLAUSIBILITY CHECKS

RC should have units of seconds. For the three required time steps, voltage should start at 5 V, remain positive, and decrease toward zero. Evaluate the exact solution at the actual times returned by your function.

OPTIONAL EXTENSION

Try dt = 1.25 tau. Describe the behavior and whether it makes physical sense.

NOTES / OBSERVATIONS

Submit your code and error results for review before requesting a solution.